PaveWatch: Autonomous Surface Deformation Monitoring

Shisir Baral[PM,1], Abhinav Dhakal[CS,1], Aalok Uprety[1], Ankit Dhakal[2], Pritam Adhikari[1], Saugat Karki[1]
Jingdao Chen[A, 3]

[PM] – Project Manager
[CS] – Chief Scientist
[A] – Faculty Advisor

[1] – Computer Engineering
[2] – Electrical Engineering
3 – Computer Science & Engineering

The Problem:
Sidewalks, roads, parking areas, and other paved surfaces gradually crack, heave, settle, and erode because of factors such as tree-root growth, freeze-thaw cycles, loading, weather, and normal wear. Because this deterioration often develops slowly, defects may go undocumented until they become large enough to create safety hazards, increase maintenance costs, or require more extensive repairs. The American Society of Civil Engineers (ASCE) gave U.S. roads a D+ in its 2025 Report Card for America's Infrastructure and estimated a national infrastructure funding gap of approximately $3.7 trillion.

Surface deterioration is also a safety concern for pedestrians. According to the National Safety Council and the National Council on Aging (NCOA), falls are the leading cause of injury-related death among older adults, resulting in more than 43,000 deaths in 2024 and approximately 3.85 million emergency-room visits in 2023, with an estimated annual economic impact of roughly $80 billion. Uneven or deteriorated walking surfaces can create trip hazards, making early identification and tracking of pavement deformation important for maintenance planning and public safety.

Current pavement inspection methods often depend on personnel visually examining large areas and documenting defects manually. This process can be slow, labor-intensive, and subject to human errors. It can also be difficult to determine whether a crack, settlement area, or other defect has changed significantly between inspections when measurements are not collected in a repeatable way. Industry reporting from DSLRpros indicates that drone-based infrastructure inspection can reduce inspection time by as much as 70% and reduce costs by approximately 40% to 60% compared with conventional inspection crews. These limitations create a need for a repeatable inspection system that can collect consistent surface measurements, compare conditions over time, and automatically identify areas that require closer attention.

Proposed Solution:
The gap between what can be measured automatically and what is still commonly inspected manually is where PaveWatch fits in. PaveWatch is a proposed autonomous unmanned aerial vehicle (UAV) system designed to survey sidewalks, roads, parking areas, and other exposed paved surfaces along repeatable flight paths. Its purpose is to detect, measure, and track surface deformation such as cracking, heaving, settlement, and erosion over multiple inspection dates.

For each survey, an operator defines the inspection area and uploads a pre-planned route. The UAV then follows that route using GPS-based navigation while a sensing payload collects high-resolution measurements of the pavement surface. After each flight, the collected data will be

offloaded to cloud servers for processing. The processing system will organize and align the new survey data, generate a georeferenced representation of the inspected surface, and compare it with historic survey data. Differences between the current and baseline measurements will be used to identify changes in surface elevation, crack development, settlement, or other measurable deformation.

The system will present the results through a user-accessible reporting interface. This allows engineers or maintenance personnel to identify where deterioration is occurring, determine how it is changing over time, and prioritize locations for further inspection or repair.

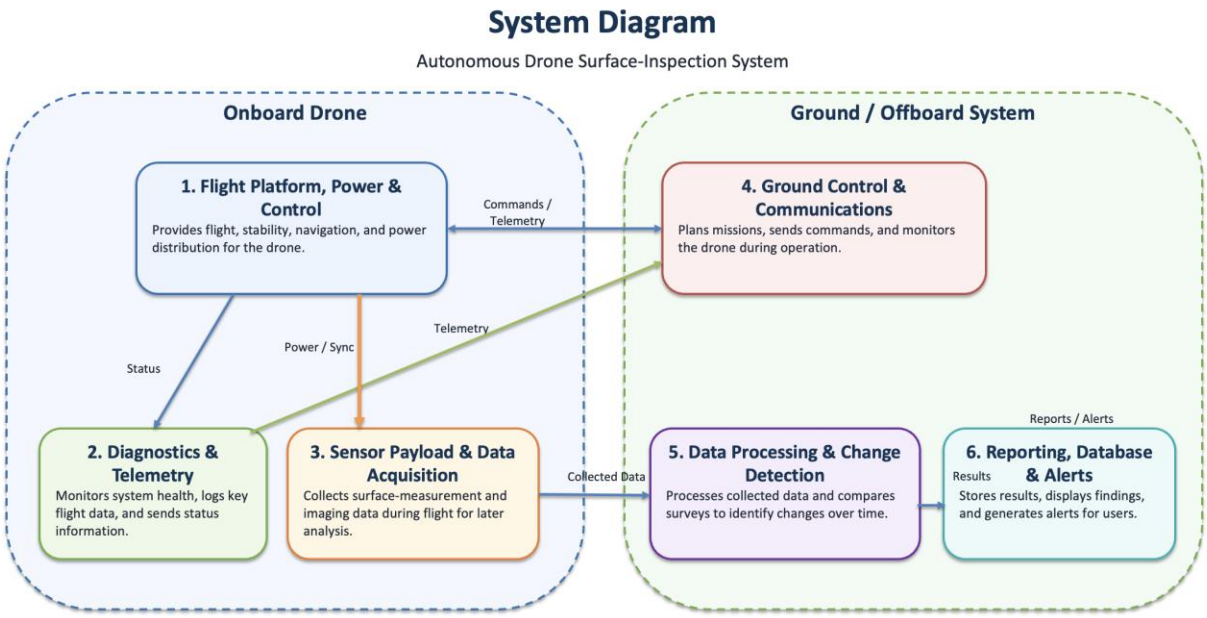
By automating repeatable data collection and comparison over time, PaveWatch is intended to reduce the amount of manual inspection required, improve consistency between inspections, provide objective records of surface condition, and help maintenance personnel identify developing pavement problems before they become more severe.

Process Diagram:



The process begins by planning the mission and defining the survey area and flight path. The drone then flies autonomously and collects sensor and GPS data. After the flight, the data is transferred to a computer or server, where it is processed and compared with the stored baseline survey. Any significant surface changes are identified and flagged. Finally, the results are displayed in a report or dashboard and stored for future inspections.

System Diagram:



The system is split into six subsystems. The Flight Platform, Power & Control subsystem handles autonomous flight, navigation, and power distribution, while the Sensor Payload & Data Acquisition subsystem collects surface and imaging data. Diagnostics & Telemetry monitors system health and sends status information to the ground station. The Ground Control & Communications subsystem is used to plan missions, send commands, and monitor the drone. After the flight, the collected data is sent to Data Processing & Change Detection, where it is analyzed and compared with previous survey data. The final results are then sent to Reporting, Database, & Alerts for storage, visualization, and notification of any detected changes.

Team Charter
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Autonomous Surface Deformation Monitoring
S206, S323

Department / Company	Advisors	
CSE	**Jingdao Chen**	**chenjingdao@cse.msstate.edu**
Expected Meeting Time & Frequency w/ Advisor(s): **Wednesday, 2:00 PM in 314 Butler Hall**		

Membership:

Project Manager		Chief Scientist	
Shisir Baral		**Abhinav Dhakal**	
SB3771	CPE	AD2432	CPE

Other Team Members			
Pritam Adhikari		**Ankit Dhakal**	
PA511	EE	AD2436	CPE
Saugat Karki		**Aalok Uprety**	
SK2088	CPE	AU160	CPE

Mode & Frequency of Communication:

We will meet for a team meeting every week, Saturdays 8:30 pm - 10:00 pm for an in-person meeting. The meetings will be held at the PMs apartment, unless specified otherwise. Anyone unable to be present will join remotely via a video call. Updates will be sent to the group chat when required.
All the shared documents will be saved and shared via OneDrive and Teams Channel. Relevant files may be shared via GitHub.
We plan to meet with our advisors once a week at the specified time.

Submission Schedule:

The team will complete internal reviews before each official course deadline.

Deliverable	Internal Review / Milestone	Deadline
Proposal/Team Charter First Draft	Team review and submission check	August 27, 2026
Proposal/Team Charter Final Draft	Complete edits	August 31, 2026
Proposal/Team Charter Final Draft	Advisor review	September 2, 2026
Proposal/Team Charter Final Draft	Final submission check	September 3, 2026
Problem Statement Peer Review Draft	Draft complete	September 2, 2026
PDR First Draft	Draft complete	September 7, 2026
PDR First Draft	Team review	September 8, 2026
PDR First Draft	Final submission check	September 10, 2026
Problem Statement First Draft	Draft complete	September 10, 2026
Problem Statement First Draft	Team review	September 12, 2026
Problem Statement First Draft	Final submission check	September 13, 2026
PDR Final Draft	Complete team and advisor review	September 14, 2026
PDR Final Draft	Freeze final submission	September 15, 2026
Problem Statement Final Draft	Complete team review	September 19, 2026
Problem Statement Final Draft	Advisor review	September 21, 2026
Problem Statement Final Draft	Final submission check	September 22, 2026
Design Requirements Specification First Draft and Peer Review	Complete draft and peer reviews	October 6, 2026
MSDR Document	Complete document	October 12, 2026
MSDR Document	Team review	October 15, 2026
MSDR Document	Finalize	October 18, 2026
Design Requirements Specification Final Draft	Advisor review	October 16, 2026
Design Requirements Specification Final Draft	Finalize	October 18, 2026

Demonstration Rubric First Draft	Team review	October 22, 2026
Approach Peer Review Draft	Complete draft and peer reviews	October 22, 2026
Approach First Draft	Team review	October 28, 2026
Approach First Draft	Finalize	October 29, 2026
Approach Final Draft	Team review	November 7, 2026
Approach Final Draft	Advisor review	November 10, 2026
Approach Final Draft	Finalize	November 12, 2026
Demonstration Rubric Final	Team review	November 18, 2026
Demonstration Rubric Final	Finalize	November 19, 2026
Final Demonstration	Working demonstration ready	November 16, 2026
Final Demonstration	Integration testing complete	November 20, 2026
Final Demonstration	Full rehearsal	November 23, 2026
Final Demonstration	Final acceptance testing complete	November 27, 2026
Final DR PP, Document, and Website	Complete drafts	November 20, 2026
Final DR PP, Document, and Website	Team review	November 27, 2026
Final DR PP, Document, and Website	Advisor review and testing	December 2, 2026

Team Roles:

Project Subsystems

1	Flight Platform, Power & Control	Ankit Dhakal (lead)	Aalok Uprety (support)
	This subsystem makes the drone fly, and supplies power to every other subsystem.		
2	Diagnostics & Telemetry	Shisir Baral (lead)	Ankit Dhakal (support)
	Monitors the drone's health, system status, faults, and real-time telemetry.		
3	Sensor Payload & Data Acquisition	Abhinav Dhakal (lead)	Saugat Karki (support)
	This subsystem collects sensor and camera data during flight and saves it for later processing.		
4	Ground Control & Communications	Pritam Adhikari (lead)	Abhinav Dhakal (support)
	This subsystem lets the operator plan missions, send position corrections, and monitor the drone from the ground.		
5	**Data Processing & Change Detection**	Aalok Uprety (lead)	Shisir Baral (support)
	This subsystem processes the collected data and finds where the surface has changed over time.		
6	**Reporting, Database & Alerts**	Saugat Karki (lead)	Pritam Adhikari (support)
	This subsystem stores results, shows them on a dashboard, and alerts staff when a problem is found.		

Potential Obstacles:

Technically: The team may disagree on technical decisions such as sensor selection, hardware platforms, software tools, or implementation approaches.
Financially: Budget limitations may create disagreements over purchasing decisions, example might be buing a prebuilt component that will make the work easier but cost more.
Logistically: Scheduling test sites, obtaining permission to operate the UAV, coordinating access to equipment, and arranging times when team members are available may cause delays.

Conflict Resolution:

For resolving conflicts, we plan to resolve them through our weekly meetings. Opposing viewpoints are laid out clearly, then everyone is asked to provide their thoughts on the matter. If the discussions fail to be fruitful, we will move to anonymous voting to decide what to do (tools like strawpoll.com may be utilized). Major decisions will require consultation with the advisor and majority support (4/6).
If the votes are tied still, the CS and PM will make the final decision for their respective fields.

Additional Comments or Concerns:

[Signatures on the Next Page.]

Signatures

Team Signatures

Signature	09/04/2026 Date	Signature	09/04/2026 Date
Shisir Baral		Abhinav Dhakal	
Project Manager		**Chief Scientist**	
Signature	09/04/2026 Date	Signature	09/04/2026 Date
Pritam Adhikari		Ankit Dhakal	
Member		**Member**	
Signature	09/04/2026 Date	Signature	09/04/2026 Date
Saugat Karki		Aalok Uprety	
Member		**Member**	

Advisor Signatures

Jing Dao Chen Digitally signed by Jing Dao Chen Date: 2026.09.05 10:35:24 -05'00'			
Signature	Date	Signature	Date
Jingdao Chen		[External Advisor's Name]	
Computer Science and Engineering		[External Advisor's Institution]	
Faculty Advisor		**External Advisor**	

References:

American Society of Civil Engineers. "2025 Report Card for America's Infrastructure." *ASCE*, 2025, infrastructurereportcard.org.

American Society of Civil Engineers. "Wastewater." *2025 Report Card for America's Infrastructure*, ASCE, 2025, infrastructurereportcard.org/cat-item/wastewater-infrastructure/.

DSLRpros. "Drone Infrastructure Inspection Solutions Guide." *DSLRpros*, 29 Aug. 2025, www.dslrpros.com/blogs/drone-trends/drone-infrastructure-inspection-for-bridges-roads-and-buildings.

National Council on Aging. "Get the Facts on Falls Prevention." *NCOA*, ncoa.org/article/get-the-facts-on-falls-prevention/.

National Safety Council. "Older Adult Falls." *Injury Facts*, NSC, 26 Mar. 2026, injuryfacts.nsc.org/home-and-community/safety-topics/older-adult-falls/.

United States, Occupational Safety and Health Administration. "Investigation into Two Workers' Deaths in Edmund Finds Arkansas Contractor Failed to Test Oxygen Level in Confined Space, Provide Rescue Equipment." *OSHA*, 7 Dec. 2022, www.osha.gov/news/newsreleases/region6/12072022.

AI Acknowledgment

Generative artificial intelligence (AI) tools were used in the preparation of this document. AI-assisted image editing was used extensively to refine the visual presentation, layout, and formatting of the **Process Diagram** and **System Diagram**. The diagrams were reviewed by the team to ensure that the final versions accurately represent the intended project workflow and system architecture.

AI tools were also used as a final-pass writing aid to review portions of the document for grammar, clarity, organization, and readability. The project concept, engineering decisions, technical content, source selection, and final submitted material were reviewed and approved by the team. The team remains responsible for the accuracy and content of the final document.